Customer Shopping Behavior Analysis

Insights from 3,900 purchases across product categories to guide strategic business decisions.

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Fields

Customer and purchase details

37

Missing Values

Review ratings imputed

Key Features

- Customer demographics (Age, Gender, Location)
- Purchase details (Category, Amount, Season)
- Shopping behavior (Discounts, Promo Codes)

Data Preparation

Python cleaning with `pandas`, PostgreSQL integration, and feature engineering including age groups and purchase frequency calculations.

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Revenue by Gender

Male Customers

$157,890 total revenue

2,652 purchases recorded

Female Customers

$75,191 total revenue

1,248 purchases recorded

Male customers generated **2.1x more revenue** than female customers in this dataset.

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Top Products by Rating

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

3.80 average rating

5

Skirt

3.78 average rating

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Shipping & Discount Analysis

Shipping Type Comparison

- **Standard:** $58.46 average
- **Express:** $60.48 average

Express shipping customers spend **$2 more** per purchase.

Discount-Dependent Products

1. Hat (50.00%)
2. Sneakers (49.66%)
3. Coat (49.07%)
4. Sweater (48.17%)
5. Pants (47.37%)

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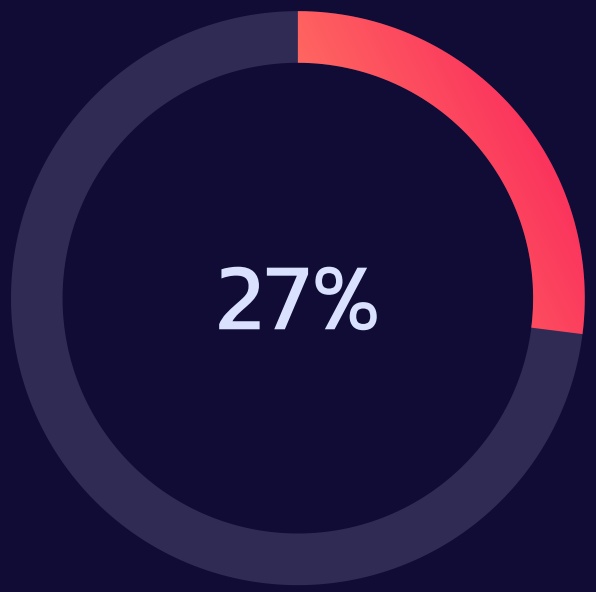
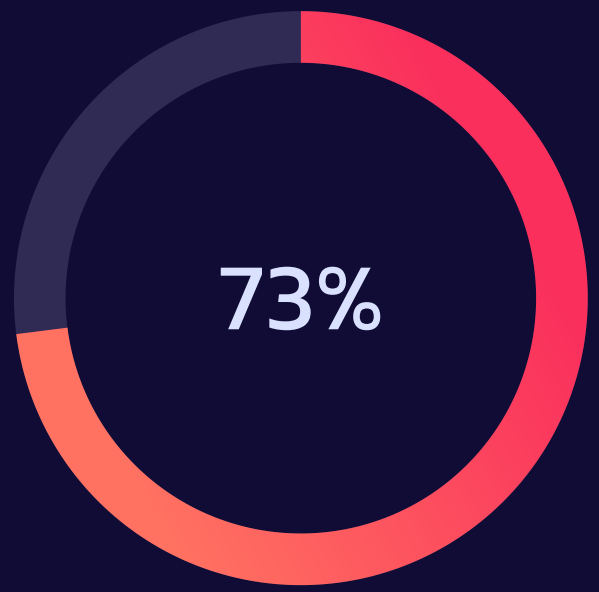
Subscription Impact



Subscribers

1,053 customers



Non-Subscribers

2,847 customers

Average Spend

- Subscribers: $59.49
- Non-Subscribers: $59.87

Total Revenue

- Subscribers: $62,645
- Non-Subscribers: $170,436

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Customer Segmentation

Loyal

3,116 customers with consistent purchases

Returning

701 customers making repeat purchases

New

83 customers with single transactions

83% of customers are classified as loyal or returning, indicating strong retention.

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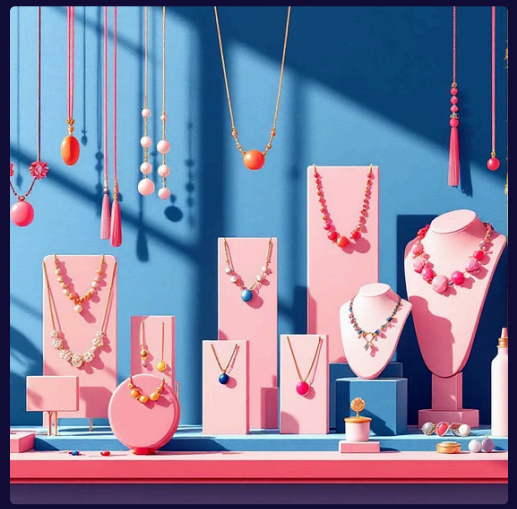
Top Products by Category



Accessories

Jewelry: 171 orders

Sunglasses: 161 orders

Belt: 161 orders

Clothing

Blouse: 171 orders

Pants: 171 orders

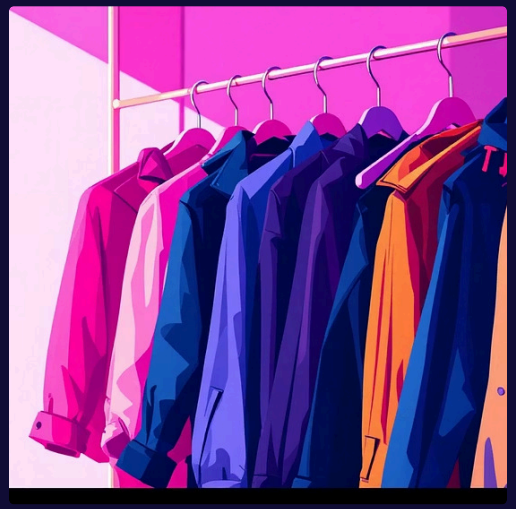
Shirt: 169 orders

Footwear

Sandals: 160 orders

Shoes: 150 orders

Sneakers: 145 orders



Outerwear

Jacket: 163 orders

Coat: 161 orders

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Revenue by Age Group

Young Adult

$62,143 total revenue

Middle-aged

$59,197 total revenue

Adult

$55,978 total revenue

Senior

$55,763 total revenue

Young adults generated the highest revenue, followed closely by middle-aged customers.

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Strategic Recommendations

01

Boost Subscriptions

Promote exclusive benefits to increase subscriber base from current 27%.

02

Enhance Loyalty Programs

Reward repeat buyers to accelerate movement into loyal customer segment.

03

Optimize Discount Policy

Balance sales boosts with margin control, especially for high-discount products.

04

Strategic Product Positioning

Highlight top-rated products (Gloves, Sandals) and best-sellers (Jewelry, Blouse) in campaigns.

05

Targeted Marketing

Focus on high-revenue age groups and express-shipping users for maximum ROI.

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